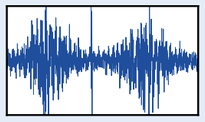
(a) Processing pipeline and model



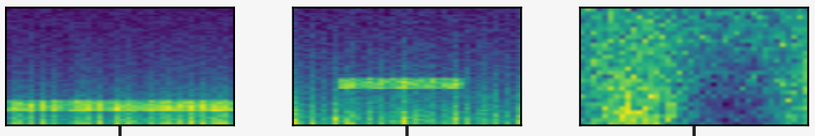
Respiratory sound recordings
ICBHI 2017 · KAUH · RespiratoryDatabase@TR
1,667 recordings, 7 classes

Pre-processing
Resampling to 4 kHz · 5 s frames, 50% overlap
Baseline-wander removal (0–1 Hz DFT bins zeroed) · max-abs normalization

Augmentation of non-COPD frames (waveform)
Time stretch 0.85–1.15, pitch shift ±3 semitones, Gaussian noise (factor 0.001–0.01), or all three
Each class raised to 9,845 frames

Time–frequency maps of each frame

Mel spectrogram	CQT	MFCC
64 bands	64 bins from 32.7 Hz	30 coefficients
64 × 39	64 × 40	30 × 40



Min–max scaling to [0, 1], resizing to 64 × 40, stacking as a **3 × 64 × 40** input

Frame-level stratified 5-fold split (after augmentation)
Test: real frames only · validation: 10% of training frames
Additional check: one patient-disjoint fold

FreqCCT (proposed model)

Tokenizer 1: conv 3×3, 3 → 64, ReLU, max-pool (3, 2, 1)

Tokenizer 2: depthwise 3×3 + pointwise 1×1, 64 → 64, ReLU, max-pool

160 tokens × 64
(16 frequency rows × 10 time columns)

+ Learned positional embedding (160 × 64)

Transformer encoder × 4
pre-LN, 4 heads, d = 64 (block in b)

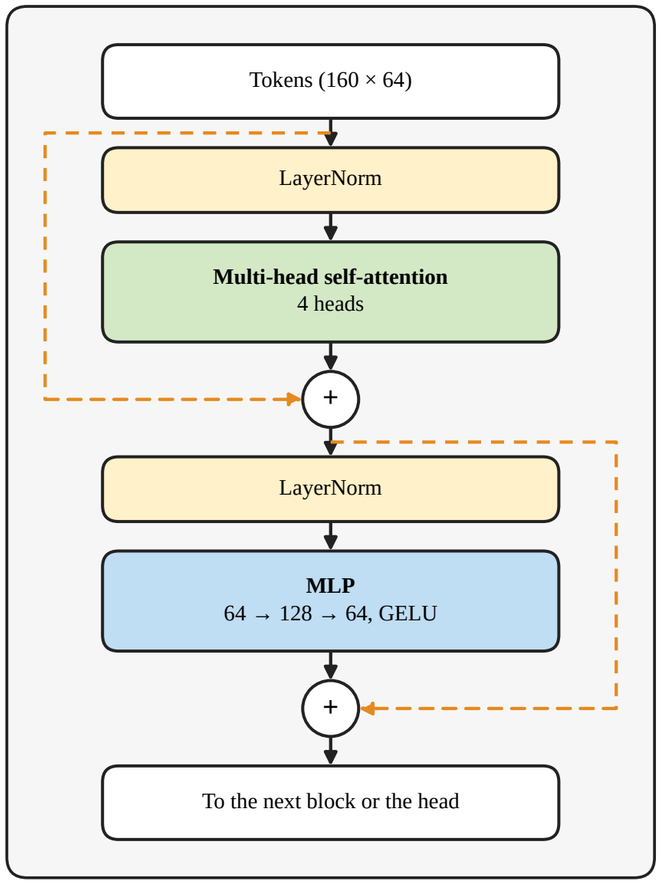
Head: LayerNorm, mean over 10 time tokens, 16 rows kept (1,024 features), linear 1,024 → 7

Changed relative to the CCT (ablation rows B–D)

Frame-level output
class probabilities, 7 classes

Recording-level output
mean of frame probabilities, arg max

(b) Pre-LN encoder block (one of four)



Dashed orange lines: residual connections

Dropout rate: 0

(c) Explanation and cost analysis

Final model
real test frames of fold 0 (n = 2,360)

Explanation maps on the 16 × 10 token grid

Grad-CAM
true class, tokenizer output

Attention rollout
attention weights of (b), 0.5 W + 0.5 I

Integrated Gradients
training-mean baseline, 64 steps

Tests of the explanations

Rollout vs Grad-CAM agreement
Spearman ρ, top-10% IoU, shuffle null

Token deletion vs random order
AUC, bootstrap over recordings

Randomization sanity check
trained vs untrained model

ICBHI pointing test
annotated crackle and wheeze cycles

Band-energy test
mel and CQT Hz axes, BH-FDR

Exact channel Shapley values
mel, CQT, MFCC; 8 coalitions

Shift and shuffle test
CCT vs FreqCCT

Evidence clips
0.5 s occlusion, original audio

Band selection and cost

Band selection
rows ranked by Grad-CAM on validation frames; models retrained with 12 or 8 of 16 rows

Cost per 5 s frame
parameters, MFLOPs (attention included), int8 weight size, peak activation, single-thread CPU time